



Differential Equations, Initial Value Problems:

Given the IVP : $y'' + ay' + by = r(t)$, $y(0) = k_0$, $y'(0) = k_1$

where a and b are constants.

$r(t)$: Input (driving force)

$y(t)$: Output (response to the input)

Let us transform the equation

$$\{s^2 Y - sy(0) - y'(0)\} + a\{sY - y(0)\} + bY = R(s)$$

$$\Rightarrow (s^2 + as + b)Y(s) = (s+a)y(0) + y'(0) + R(s)$$

$$\Rightarrow Y(s) = \frac{(s+a)y(0) + y'(0) + R(s)}{(s^2 + as + b)}$$

Let us define the Transfer function as

$$Q(s) = \frac{1}{s^2 + as + b} = \frac{1}{(s + \frac{1}{2}a)^2 + b^2 - \frac{1}{4}a^2}$$

$$\text{Then we have } Y(s) = [(s+a)y(0) + y'(0)]Q(s) + R(s)Q(s)$$

If the system is originally at rest, then

$$y(0) = 0, y'(0) = 0 \quad \text{and} \quad \boxed{Y(s) = Q(s)R(s)}$$

$$\left(\begin{array}{l} \text{Note: Transfer function } Q = \frac{Y}{R} = \frac{\mathcal{L}(\text{output})}{\mathcal{L}(\text{Input})} \\ \text{This explains the name } Q. \\ Q \text{ depends neither on } r(t) \text{ nor on the initial} \\ \text{conditions (but only on } a \text{ and } b) \end{array} \right)$$

To recover y we need to perform $\mathcal{L}^{-1}$ on Y

$$\text{Thus } y = \mathcal{L}^{-1}\{Y\}$$

Ex: Solve $y'' + 9y = 10e^{-t}$, $y(0) = 0$, $y'(0) = 0$

Now taking Laplace transform, we get

$$\{s^2 Y - s y(0) - y'(0)\} + 9Y = \frac{10}{s+1}$$

$$\Rightarrow (s^2 + 9)Y = \frac{10}{s+1}$$

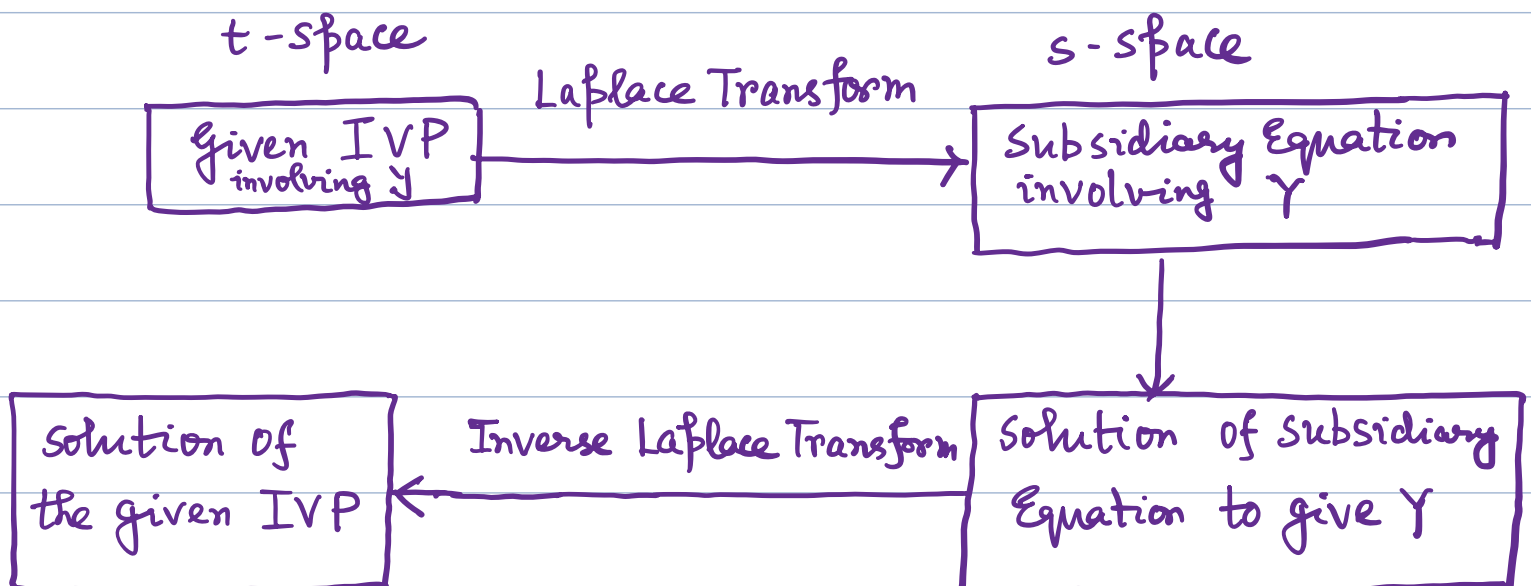
$$\Rightarrow Y = \frac{10}{(s+1)(s^2+9)}$$

$$\Rightarrow Y = \frac{1}{s+1} - \frac{s-1}{s^2+9}$$

$$\text{So, } y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s+1} - \frac{s-1}{s^2+9} \right\}$$
$$= \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} - \mathcal{L}^{-1} \left\{ \frac{s}{s^2+9} \right\} + \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{3}{s^2+9} \right\}$$

$$\boxed{y(t) = e^{-t} - \cos(3t) + \frac{1}{3} \sin(3t)}$$

Summary of the method:



Ex: $y'' - y' - 6y = 0$, $y(0) = 6$, $y'(0) = 13$

Taking Laplace Transform we get

$$\{s^2 Y - s y(0) - y'(0)\} - \{s Y - y(0)\} - 6 Y = 0$$

$$\Rightarrow \{s^2 Y - 6s - 13\} - \{s Y - 6\} - 6 Y = 0$$

$$\Rightarrow (s^2 - s - 6) Y = 6s + 7$$

$$\Rightarrow (s-3)(s+2) Y = 6s + 7$$

$$\Rightarrow Y = \frac{6s+7}{(s-3)(s+2)} = \frac{5}{s-3} + \frac{1}{s+2}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{5}{s-3}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$\Rightarrow \boxed{y(t) = 5e^{3t} + e^{-2t}}$$

(can be done directly
also by using
auxiliary equation)

Ex: $y'' - 2y' - 3y = 0$ $y(1) = -3$, $y'(1) = -17$

Problem: The initial values are not given at $t=0$

Let $x = t - 1$ (so $t=1 \Rightarrow x=0$)

$$\Rightarrow t = x + 1$$

Substituting $t = x + 1$ in y , we get a function of x

let us call it $\tilde{y}(x) = y(t) (= y(x+1))$

Now $y' = \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = \frac{dy}{dx} = \frac{d\tilde{y}}{dx}$

$$\text{and } y'' = \frac{d(y')}{dt} = \frac{d\left(\frac{d\tilde{y}}{dx}\right)}{dt} = \frac{d\left(\frac{d\tilde{y}}{dx}\right)}{dx} \frac{dx}{dt} = \frac{d\left(\frac{d\tilde{y}}{dx}\right)}{dx} = \frac{d^2\tilde{y}}{dx^2}$$

So, the IVP becomes

$$\frac{d^2\tilde{y}}{dx^2} - 2\frac{d\tilde{y}}{dx} - 3\tilde{y} = 0, \quad \tilde{y}(0) = -3, \quad \left.\frac{d\tilde{y}}{dx}\right|_{x=0} = -17$$

Taking Laplace Transform we get

$$\left\{s^2\tilde{Y} - s\tilde{y}(0) - \left.\frac{d\tilde{y}}{dx}\right|_{x=0}\right\} - 2\{s\tilde{Y} - \tilde{y}(0)\} - 3\tilde{Y} = 0$$

$$\Rightarrow (s^2 - 2s - 3)\tilde{Y} + 3s + 17 + 2(-3) = 0$$

$$\Rightarrow (s^2 - 2s - 3)\tilde{Y} = -3s - 11$$

$$\Rightarrow (s-3)(s+1)\tilde{Y} = -3s - 11$$

$$\Rightarrow \tilde{Y} = \frac{-3s-11}{(s-3)(s+1)} = \frac{2}{s+1} - \frac{5}{s-3}$$

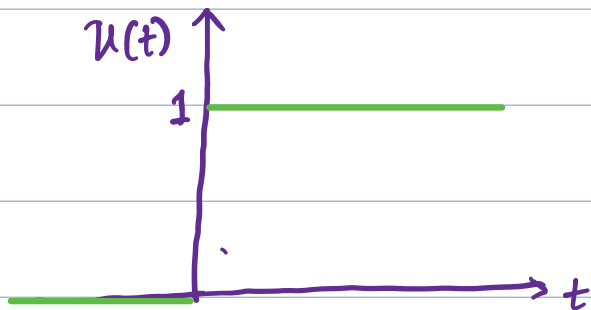
$$\Rightarrow \tilde{y} = \mathcal{L}^{-1}\left\{\frac{2}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{5}{s-3}\right\}$$

$$\Rightarrow \tilde{y} = 2e^{-x} - 5e^{3x} \quad \left(\text{This is the solution of the shifted problem}\right)$$

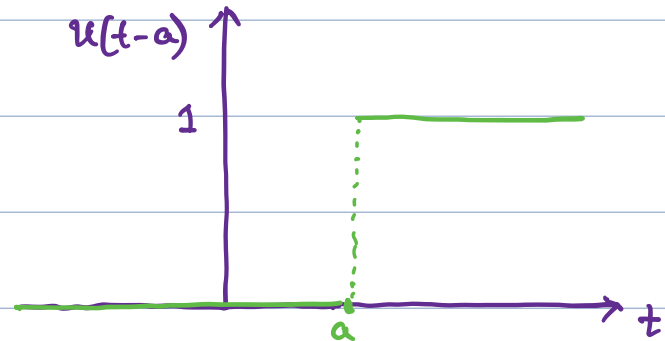
Hence $y = 2e^{-(t-1)} - 5e^{3(t-1)}$

Unit step function (Heaviside function):

$$u(t-a) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t > a \end{cases} \quad \text{where } a \in \mathbb{R}$$



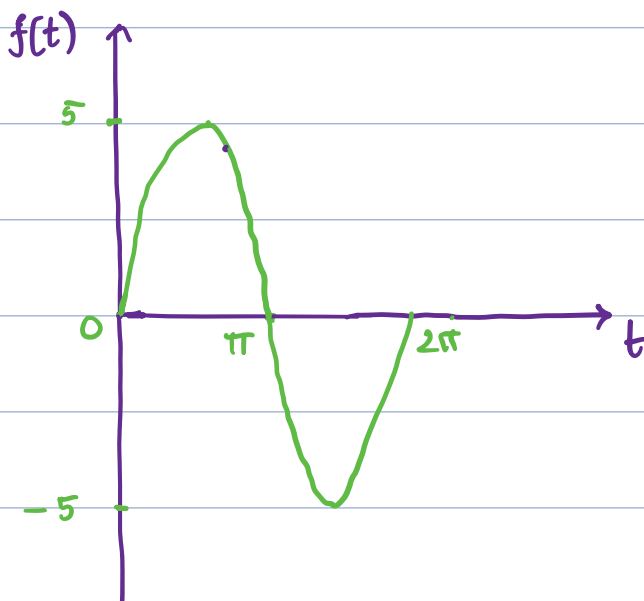
Unit Step function $u(t)$



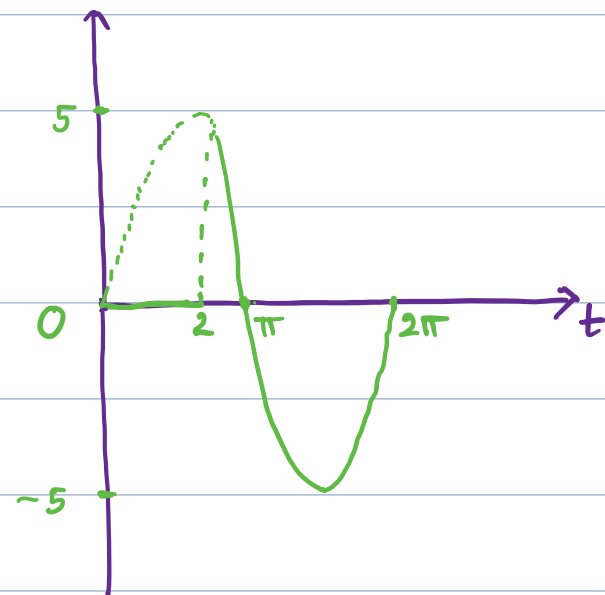
Unit Step function $u(t-a)$

$$\begin{aligned} \mathcal{L}\{u(t-a)\} &= \int_0^{\infty} e^{-st} u(t-a) dt = \int_a^{\infty} e^{-st} dt \\ &= \left. \frac{e^{-st}}{-s} \right|_a^{\infty} = \frac{e^{-as}}{s} \end{aligned}$$

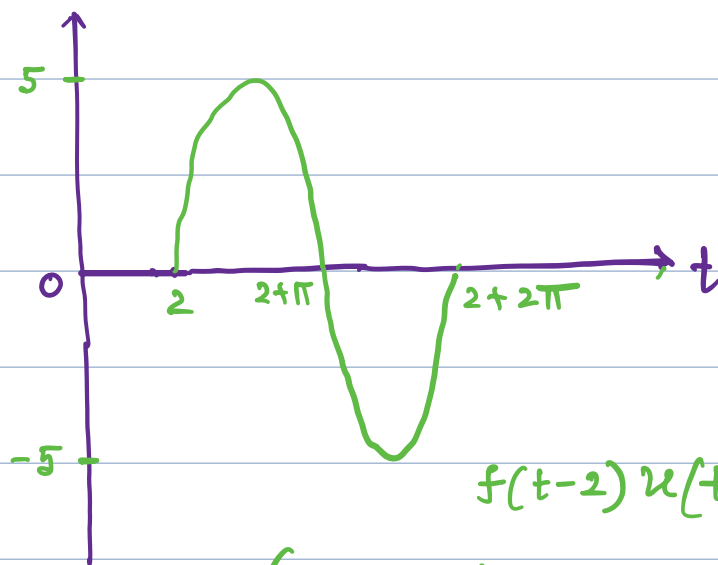
Ex:



$f(t) = 5 \sin t$
(Given function)

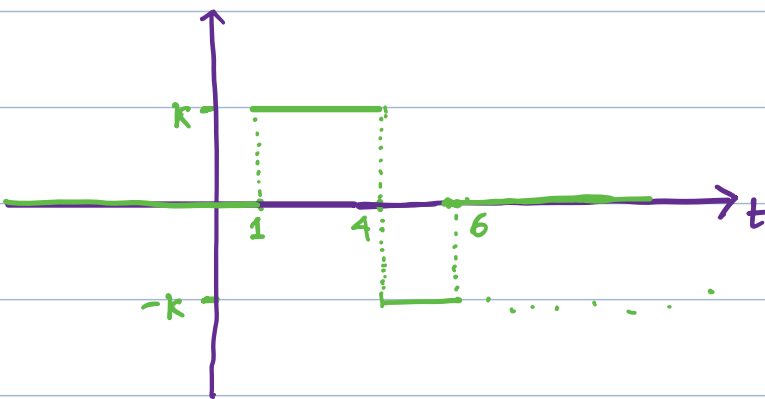


$f(t)u(t-2)$
(switching off and on)

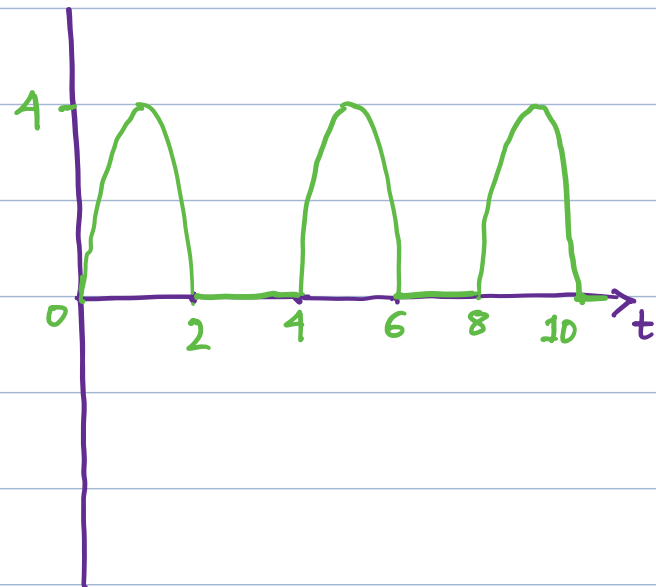


(Shift (or translate) $f(t)$ by 2)

Ex:



$$k[u(t-1) - 2u(t-4) + u(t-6)]$$



$$4\sin\left(\frac{\pi}{2}t\right)[u(t) - u(t-2) + u(t-4) - \dots]$$

Use of Unit Step functions

Time Shifting (t-shifting) Theorem:

If $f(t)$ has the Laplace transform $F(s)$, then

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

$$\mathcal{L}\{f(t)u(t-a)\} = e^{-as}\mathcal{L}\{f(t+a)\}$$

Proof: Note that $f(t-a)u(t-a) = \begin{cases} 0 & \text{if } t < a \\ f(t-a) & \text{if } t > a \end{cases}$

$$\text{So, } \mathcal{L}\{f(t-a)u(t-a)\}(s) = \int_0^{\infty} e^{-st} f(t-a)u(t-a) dt$$

$$= \int_a^{\infty} e^{-st} f(t-a) dt \quad \begin{array}{l} \text{Let } x = t-a \text{ then } dx = dt \\ \text{and } t = x+a \end{array}$$

$$= \int_0^{\infty} e^{-s(x+a)} f(x) dx = e^{-as} \int_0^{\infty} e^{-sx} f(x) dx = \boxed{e^{-as} F(s)}$$

$$\mathcal{L}^{-1}(e^{-as} F(s)) = f(t-a)u(t-a) = \boxed{e^{-as} \mathcal{L}\{f\}(s)}$$

Now $f(t)u(t-a) = \begin{cases} 0 & \text{if } t < a \\ f(t) & \text{if } t > a \end{cases}$

$$\text{So, } \mathcal{L}\{f(t)u(t-a)\}(s) = \int_0^{\infty} e^{-st} f(t)u(t-a) dt$$

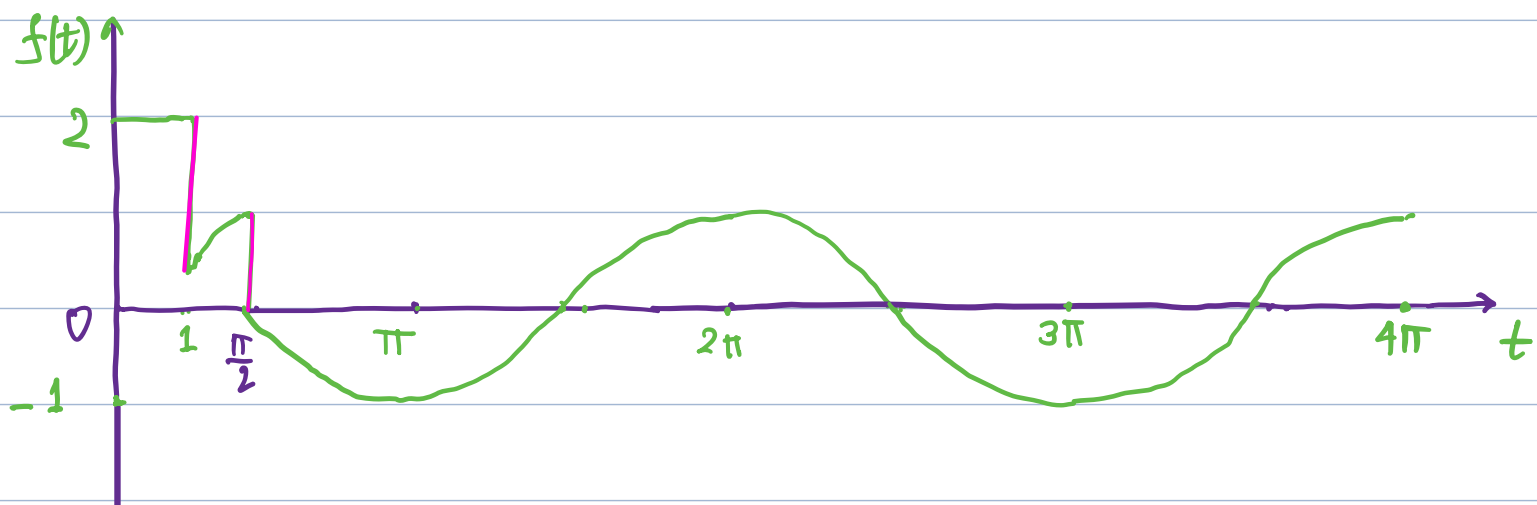
$$= \int_a^{\infty} e^{-st} f(t) dt \quad \text{Let } x = t-a \text{ then } dx = dt$$

$$= \int_0^{\infty} e^{-s(x+a)} f(x+a) dx = e^{-as} \int_0^{\infty} e^{-sx} f(x+a) dx$$

$$= e^{-as} \mathcal{L}\{f(x+a)\}(s) = e^{-as} \mathcal{L}\{f(t+a)\}(s)$$

Ex: Compute the Laplace Transform of the function:

$$\left. \begin{aligned} f(t) &= 2 \quad \text{for } 0 < t < 1 \\ &\frac{1}{2}t^2 \quad \text{for } 1 < t < \frac{\pi}{2} \\ &\cos(t) \quad \text{for } t > \frac{\pi}{2} \end{aligned} \right\}$$



$$\begin{aligned} \text{Now } f(t) &= 2 \left[1 - u(t-1) \right] + \frac{1}{2}t^2 \left[u(t-1) - u\left(t - \frac{\pi}{2}\right) \right] \\ &\quad + \cos(t) u\left(t - \frac{\pi}{2}\right) \end{aligned}$$

$$\text{Now } \mathcal{L} \left\{ 2 \left[1 - u(t-1) \right] \right\} = 2 \left(\frac{1}{s} - \frac{e^{-s}}{s} \right) = 2 \left(\frac{1 - e^{-s}}{s} \right)$$

$$\begin{aligned} \mathcal{L} \left[\frac{1}{2}t^2 u(t-1) \right] &= \mathcal{L} \left[\left\{ \frac{1}{2}(t-1)^2 + (t-1) + \frac{1}{2} \right\} u(t-1) \right] \\ &= \left[\frac{2}{2s^3} + \frac{1}{s^2} + \frac{1}{2s} \right] e^{-s} \end{aligned}$$

$$= \left[\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s} \right] e^{-s}$$

$$\mathcal{L} \left[\frac{1}{2} t^2 u \left(t - \frac{\pi}{2} \right) \right] = \mathcal{L} \left[\left\{ \frac{1}{2} \left(t - \frac{\pi}{2} \right)^2 + \frac{\pi}{2} \left(t - \frac{\pi}{2} \right) + \frac{\pi^2}{8} \right\} \times u \left(t - \frac{\pi}{2} \right) \right]$$

$$= \left[\frac{1}{2} \frac{2}{s^3} + \frac{\pi}{2} \frac{1}{s^2} + \frac{\pi^2}{8} \frac{1}{s} \right] e^{-\frac{\pi}{2}s}$$

$$= \left[\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s} \right] e^{-\frac{\pi}{2}s}$$

$$\mathcal{L} \left\{ \cos(t) u \left(t - \frac{\pi}{2} \right) \right\} = \mathcal{L} \left\{ -\sin \left(t - \frac{\pi}{2} \right) u \left(t - \frac{\pi}{2} \right) \right\}$$

$$= -\frac{1}{s^2+1} e^{-\frac{\pi}{2}s}$$

Combining

$$\mathcal{L}\{f\} = \frac{2}{s} - \frac{2}{s} e^{-s} + \left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s} \right) e^{-s}$$

$$- \left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s} \right) e^{-\frac{\pi}{2}s} - \frac{1}{s^2+1} e^{-\frac{\pi}{2}s}$$

Note: Another way to calculate

$$\mathcal{L} \left\{ \frac{1}{2} t^2 u(t-1) \right\} = e^{-s} \mathcal{L} \left\{ \frac{1}{2} (t+1)^2 \right\}$$

$$= e^{-s} \mathcal{L} \left\{ \frac{1}{2} t^2 + t + \frac{1}{2} \right\}$$

$$= e^{-s} \left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s} \right)$$

$$\mathcal{L} \left\{ \frac{1}{2} t^2 u \left(t - \frac{\pi}{2} \right) \right\} = e^{-\frac{\pi}{2}s} \mathcal{L} \left\{ \frac{1}{2} \left(t + \frac{\pi}{2} \right)^2 \right\}$$

$$= e^{-\frac{\pi}{2}s} \mathcal{L} \left\{ \frac{1}{2} t^2 + \frac{\pi}{2} t + \frac{\pi^2}{8} \right\}$$

$$= e^{-\frac{\pi}{2}s} \left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s} \right)$$

$$\text{and } \mathcal{L} \left\{ \cos(t) u \left(t - \frac{\pi}{2} \right) \right\} = e^{-\frac{\pi}{2}s} \mathcal{L} \left\{ \cos \left(t + \frac{\pi}{2} \right) \right\}$$

$$= e^{-\frac{\pi}{2}s} \mathcal{L} \left\{ (-\sin t) \right\} = - \frac{e^{-\frac{\pi}{2}s}}{s^2 + 1}$$

Combining we get the expression for
 $\mathcal{L} \{ f \}$

Ex: Find the inverse transform of

$$F(s) = \frac{e^{-2\pi s} - e^{-3\pi s}}{s^2 + 1}$$

$$\text{Note that } \mathcal{L}^{-1} \left(\frac{1}{s^2 + 1} \right) = \frac{\sin t}{1} = \sin t$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{e^{-2\pi s}}{s^2+1}\right\} = \frac{\sin(t-2\pi)}{1} \mathcal{U}(t-2\pi) \quad \left(\text{By the second shifting theorem}\right)$$

$$= \sin t \mathcal{U}(t-2\pi)$$

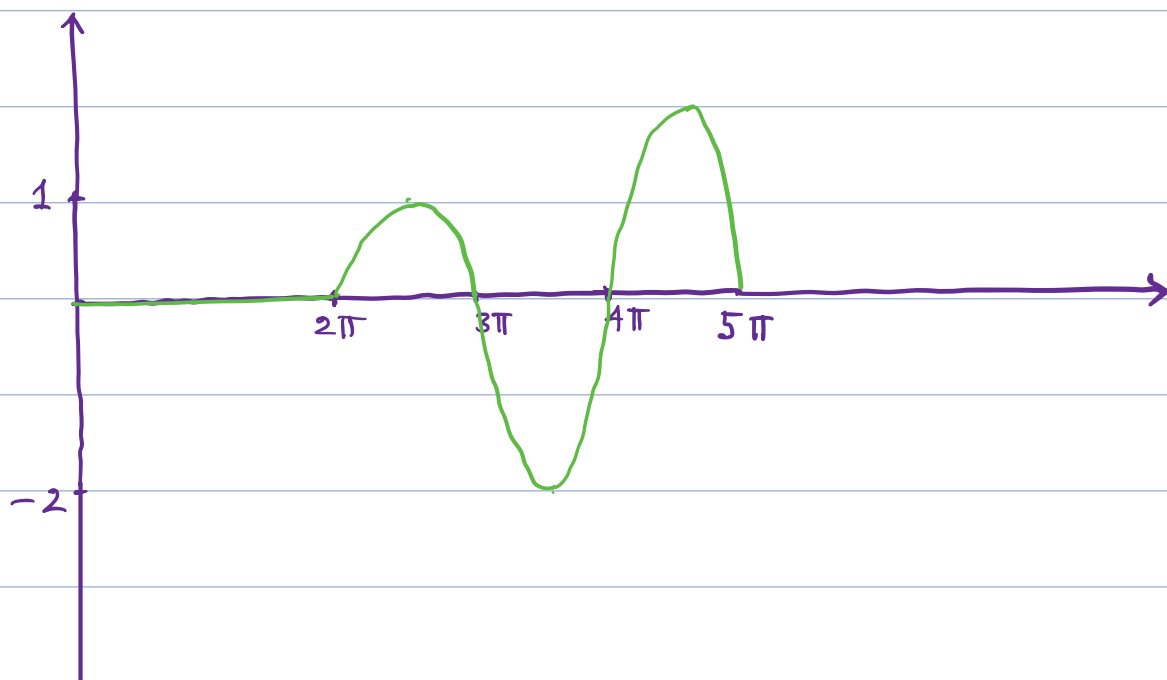
Similarly $\mathcal{L}^{-1}\left\{\frac{e^{-3\pi s}}{s^2+1}\right\} = \frac{\sin(t-3\pi)}{1} \mathcal{U}(t-3\pi)$ (second shifting theorem)

$$= \sin(t-\pi) \mathcal{U}(t-3\pi) = -\sin t \mathcal{U}(t-3\pi)$$

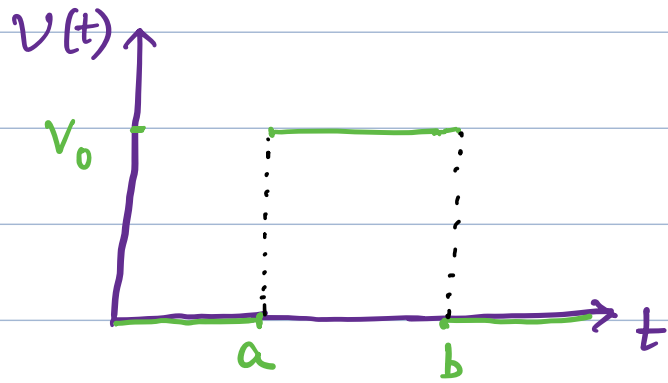
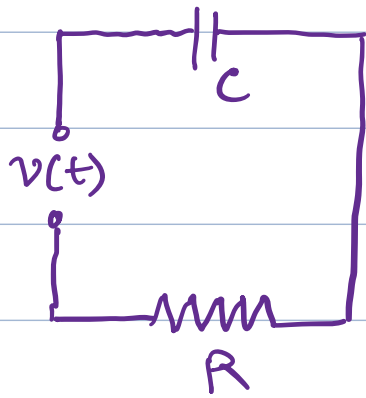
$$\text{So, } \mathcal{L}^{-1}\{F(s)\} = \sin t \mathcal{U}(t-2\pi) - (-\sin t) \mathcal{U}(t-3\pi)$$

$$= \sin t [\mathcal{U}(t-2\pi) + \mathcal{U}(t-3\pi)]$$

$$\text{or } \mathcal{L}^{-1}\{F(s)\} = \left. \begin{array}{l} 0 \text{ if } t \leq 2\pi \\ \sin t \text{ if } 2\pi < t < 3\pi \\ 2\sin t \text{ if } t \geq 3\pi \end{array} \right\}$$



Ex: Find the current $i(t)$ in the RC-circuit if a single rectangular wave with voltage V_0 is applied.



$$Ri(t) + \frac{q(t)}{C} = Ri(t) + \frac{1}{C} \int_0^t i(\tau) d\tau = V_0 [u(t-a) - u(t-b)]$$

Using Laplace transform

$$RI + \frac{1}{C} \frac{1}{s} I = V_0 \left(\frac{e^{-as}}{s} - \frac{e^{-bs}}{s} \right)$$

$$\Rightarrow I \left(R + \frac{1}{Cs} \right) = V_0 \frac{1}{s} (e^{-as} - e^{-bs})$$

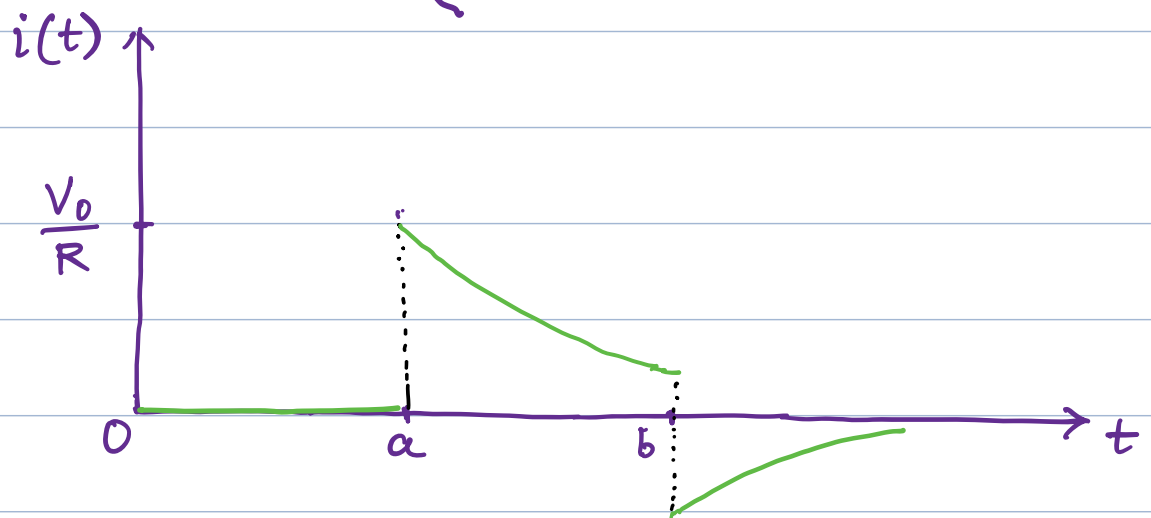
$$\Rightarrow I = \frac{Cs}{Rcs + 1} V_0 \frac{1}{s} (e^{-as} - e^{-bs})$$

$$\Rightarrow I = \frac{eV_0}{R(s + \frac{1}{RC})} (e^{-as} - e^{-bs})$$

$$\Rightarrow I = \frac{V_0}{R} \frac{1}{s + \frac{1}{RC}} (e^{-as} - e^{-bs})$$

$$\text{Now } \mathcal{L}^{-1} \left\{ \frac{V_0}{R} \cdot \frac{1}{s + \frac{1}{RC}} \right\} = \frac{V_0}{R} e^{-\frac{t}{RC}}$$

$$\begin{aligned} \text{Hence } i &= \mathcal{L}^{-1} \left\{ \frac{V_0}{R} \frac{1}{s + \frac{1}{RC}} (e^{-as} - e^{-bs}) \right\} \\ &= \frac{V_0}{R} \left\{ e^{-\frac{t-a}{RC}} u(t-a) - e^{-\frac{(t-b)}{RC}} u(t-b) \right\} \\ &\quad \text{(Using } t\text{-shifting theorem)} \end{aligned}$$



$$\begin{aligned} \text{Now } i(t) &= 0 \quad \text{if } t < a \\ &= k_1 e^{-t/RC} \quad \text{if } a < t < b \\ &= (k_1 - k_2) e^{-t/RC} \quad \text{for } t > b \end{aligned}$$

$$\text{where } k_1 = V_0 e^{a/RC}, \quad k_2 = V_0 e^{b/RC} \quad (\text{Note that } k_1 < k_2)$$

